

## SPECIFICATIONS

|            |                          |
|------------|--------------------------|
| Messrs.    | FURUNO ELECTRIC CO.,LTD. |
| Item       | Magnetic Buzzer          |
| Model Name | TE-8540BYT3H2            |
| Issue Date | Feb., 15, 2013           |
| Spec №     | TPI-110310-13B15         |

Approved by

### SUPPLIER

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| Magnetic Buzzer | MODEL : TE-8540BYT3H2 | Spec. No | TPI-110310-13B15 |

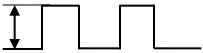
Revision List

| No | Version | Page  | Item      | Note                                                           |
|----|---------|-------|-----------|----------------------------------------------------------------|
| 1  | 00      |       |           | First issue                                                    |
| 2  | 01      | 5 /16 | Appending | Model and manufacturer name of the<br>SPL meter (Aug.,05.2013) |
| 3  |         |       |           |                                                                |
| 4  |         |       |           |                                                                |
| 5  |         |       |           |                                                                |
| 6  |         |       |           |                                                                |
| 7  |         |       |           |                                                                |
| 8  |         |       |           |                                                                |
| 9  |         |       |           |                                                                |
| 10 |         |       |           |                                                                |
|    |         |       |           |                                                                |
|    |         |       |           |                                                                |

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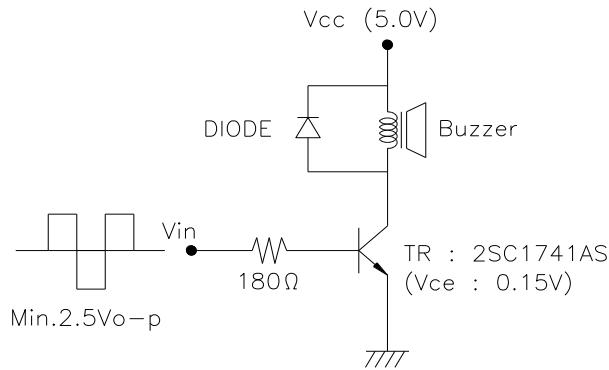
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| Magnetic Buzzer                                                                                                                                                                                                                                                            | MODEL : TE-8540BYT3H2 | Spec. No                        | TPI-110310-13B15                                                                                                                                                                                    |
| <div>1. Application</div> <div>This specification is applied to Techtuit Magnetic Buzzer unit Model TE-8540BYT3H2.</div> <div>2. Electrical Characteristics</div>                                                                                                          |                       |                                 |                                                                                                                                                                                                     |
| No                                                                                                                                                                                                                                                                         | Item                  | Specification                   | Condition                                                                                                                                                                                           |
| 2-1                                                                                                                                                                                                                                                                        | Rated voltage         | 5.0V                            | V = DC Voltage                                                                                                                                                                                      |
| 2-2                                                                                                                                                                                                                                                                        | Operation voltage     | 2.0 ~ 6.0V                      |                                                                                                                                                                                                     |
| 2-3                                                                                                                                                                                                                                                                        | Test Input Signal     | Min. 2.5Vo-p                    | Vo-p                                                                                                              |
| 2-4                                                                                                                                                                                                                                                                        | Rated frequency       | 3,950Hz                         | <ul style="list-style-type: none"><li>Standard State, Standard Drive Circuit.</li><li>Rated voltage, Distance at 10cm(A-range)</li><li>3,950Hz, Squarewave 1/2 Duty</li><li>Sounder Alone</li></ul> |
| 2-5                                                                                                                                                                                                                                                                        | Sound pressure level  | Min.100.0dB                     |                                                                                                                                                                                                     |
| 2-6                                                                                                                                                                                                                                                                        | Rated Current         | MAX. 120.0mA                    |                                                                                                                                                                                                     |
| 2-7                                                                                                                                                                                                                                                                        | Peak Current          | MAX. 385.0mA                    |                                                                                                                                                                                                     |
| 2-8                                                                                                                                                                                                                                                                        | Coil resistance       | 16±3.0Ω                         | * Basic state                                                                                                                                                                                       |
| 2-9                                                                                                                                                                                                                                                                        | Temperature range     | 1) Operating temp. -30 ~ +85 °C |                                                                                                                                                                                                     |
|                                                                                                                                                                                                                                                                            |                       | 2) Storage temp. -40 ~ +85 °C   |                                                                                                                                                                                                     |
| 2-10                                                                                                                                                                                                                                                                       | Mass                  | 0.45g                           |                                                                                                                                                                                                     |
| <div>• Standard State : Ordinary Temp(15~35°C), Humidity(25~85%RH), Air pressure(860~1,060hPa)<br/>In case of doubtful judgment, the test is re-performed under Basic State.</div> <div>•Basic State : Temp.(20±2°C), Humidity(60~70%RH), Air pressure(860~1,060hPa)</div> |                       |                                 |                                                                                                                                                                                                     |

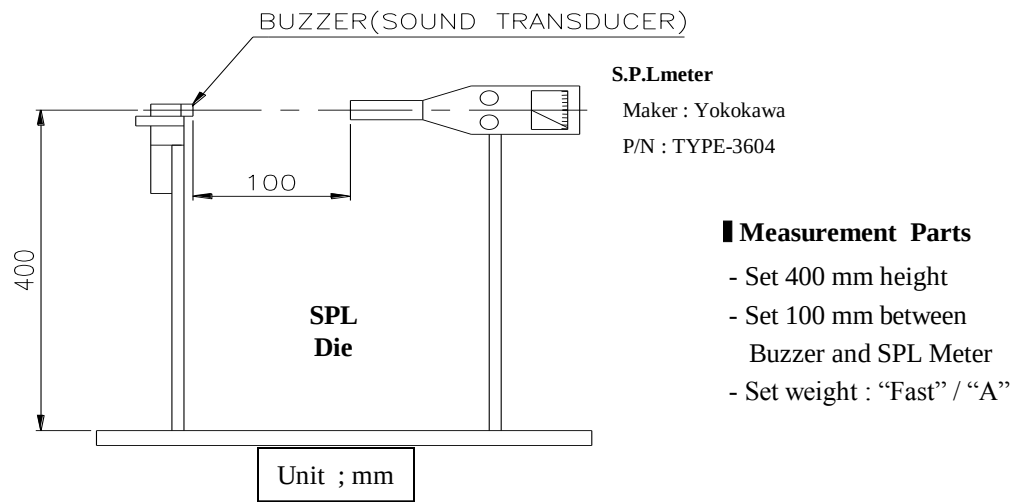
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3. Measuring Condition

3-1. Test Driver Circuit



3-2. Test Fixture



■ Measurement Parts

- Set 400 mm height
- Set 100 mm between Buzzer and SPL Meter
- Set weight : “Fast” / “A”

3-3. Normal Test Condition

| No    | Item        | Test Condition  | Notes |
|-------|-------------|-----------------|-------|
| 3-3-1 | Temperature | +5 ~ +35℃       |       |
| 3-3-2 | Humidity    | 45 ~ 85 %       |       |
| 3-3-3 | Atmosphere  | 860 ~ 1,060 hPa |       |

3-4. Basic Test Condition

| No    | Item        | Test Condition  | Notes |
|-------|-------------|-----------------|-------|
| 3-4-1 | Temperature | +20 ± 2℃        |       |
| 3-4-2 | Humidity    | 60 ~ 70 %       |       |
| 3-4-3 | Atmosphere  | 860 ~ 1,060 hPa |       |

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4. Reliability Test

| No  | Item                        | Method of test                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4-1 | Heat Resistance             | Storage in +85 °C test box for 96 hours then expose to the room temperature for 2 hours.                                                                                                                                                                                                                                                                                                        |
| 4-2 | Cold Resistance             | Storage in -40 °C test box for 96 hours then expose to the room temperature for 2 hours.                                                                                                                                                                                                                                                                                                        |
| 4-3 | Life test in the room temp. | Operate sound transducer continuously for 1000 hours with applying 5.0V, square wave, 1/2 duty, 3950Hz, using the test circuit as per attached drawing.                                                                                                                                                                                                                                         |
| 4-4 | Temperature Cycle           | <div> <div> <div> <div> <div>-30°C+0/-3</div> <div>30min</div> </div> <div> <div>+20°C+10/-5</div> <div>15min</div> </div> <div> <div>+70°C+3/-0</div> <div>30min</div> </div> <div> <div>+20°C+10/-5</div> <div>15min</div> </div> </div> <div>1 cycle : 90 min</div> </div> <p>Make this test for 5 cycle without applying power, then expose to the room temperature for 2 hours.</p> </div> |

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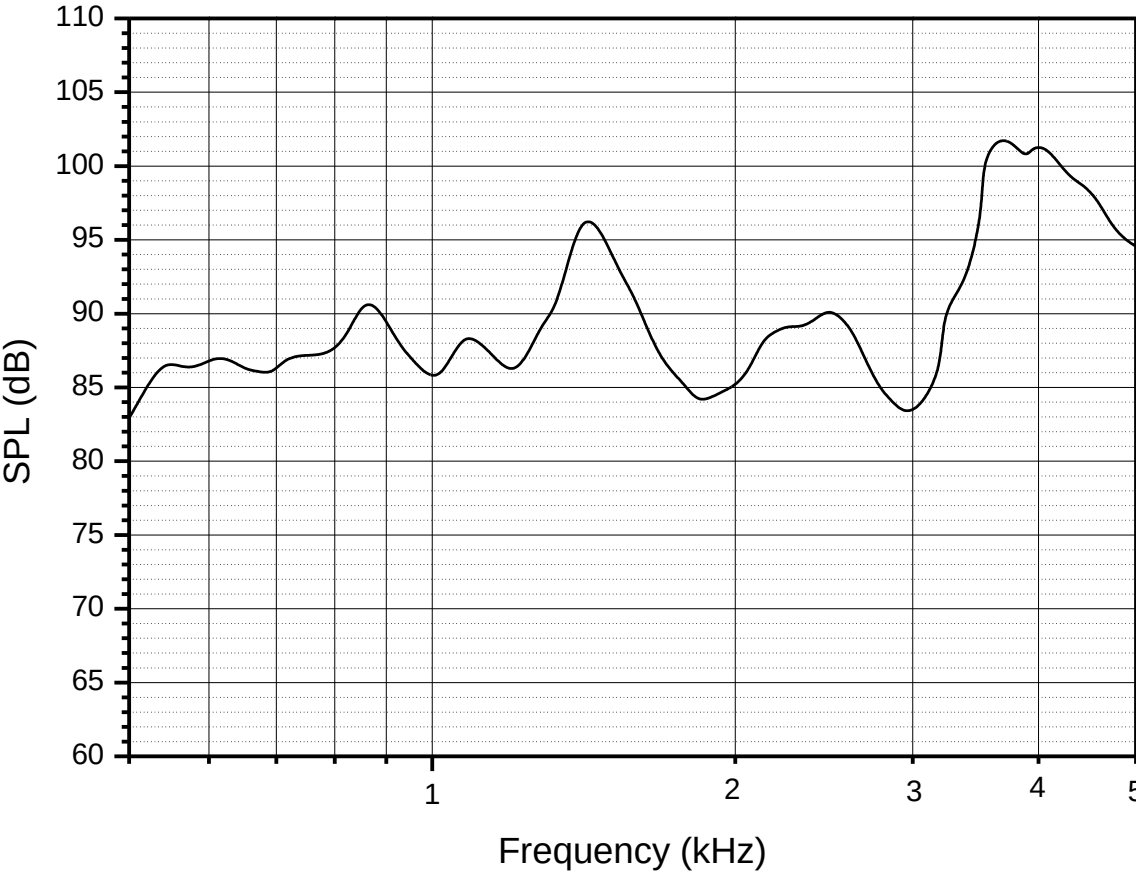
4. Reliability Test

| No  | Item                      | Method of the test                                                                                                                                                                                                                              |
|-----|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4-5 | Temp./Humidity Resistance | <div> <p> +25°C<br/>95% RH<br/>5Hr    0.5Hr    6Hr    0.5Hr<br/> 1 cycle : 12 Hr </p> </div> <p>Make this test for 10 cycles without applying power, then expose to the room temperature for 2 hours.</p>                                       |
| 4-6 | Vibration test            | <div> <p> 9.3G    p-p 1.53mm<br/> 0.3G<br/> 10    55Hz </p> </div> <p>Make this test for the directions of X, Y and Z.<br/>For 2 hours each (total 6 hours).<br/>To-and-fro sweep time is 1 minute</p> <p>Dimension of PCB : 60X90X1.6t(mm)</p> |
| 4-7 | Drop test                 | <p>Fix sound transducer on the P.C.B with Polyester film(t=0.2mm, gross weight:100g)<br/>And the height of 120cm to the direction Of six surface for 3 cycle</p>                                                                                |
| 4-8 | Solderability             | <p>The part shall be dipped into the soldering bath conditioned as follows;</p> <ul style="list-style-type: none"> <li>- Solder temp. : +235±5°C</li> <li>- Soaking time : 2.0±0.5sec.</li> </ul>                                               |

See attached file

|                 |                       |          |                  |
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6. Frequency Response

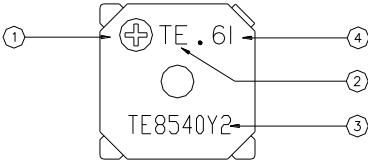


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7. Soldering Condition

| No  | Item             | Test Condition                                                                                                                                                                                                                                                                                                                                                          |
|-----|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-1 | Reflow Soldering | <p>- Recommendable reflow soldering condition is as follows.<br/> Note 1; It is requested that reflow soldering should be executed after heat of product goes down to normal temperature.<br/> Note 2; Peak reflow temperature of +260℃, with a maximum duration of 60 sec. between +220℃ and 260℃</p> <p style="text-align: center;">* Reflow profile of lead-free</p> |
| 7-2 | Manual Soldering | Soldering iron temperature : +350℃ within 5 sec                                                                                                                                                                                                                                                                                                                         |

8. Marking Specifications



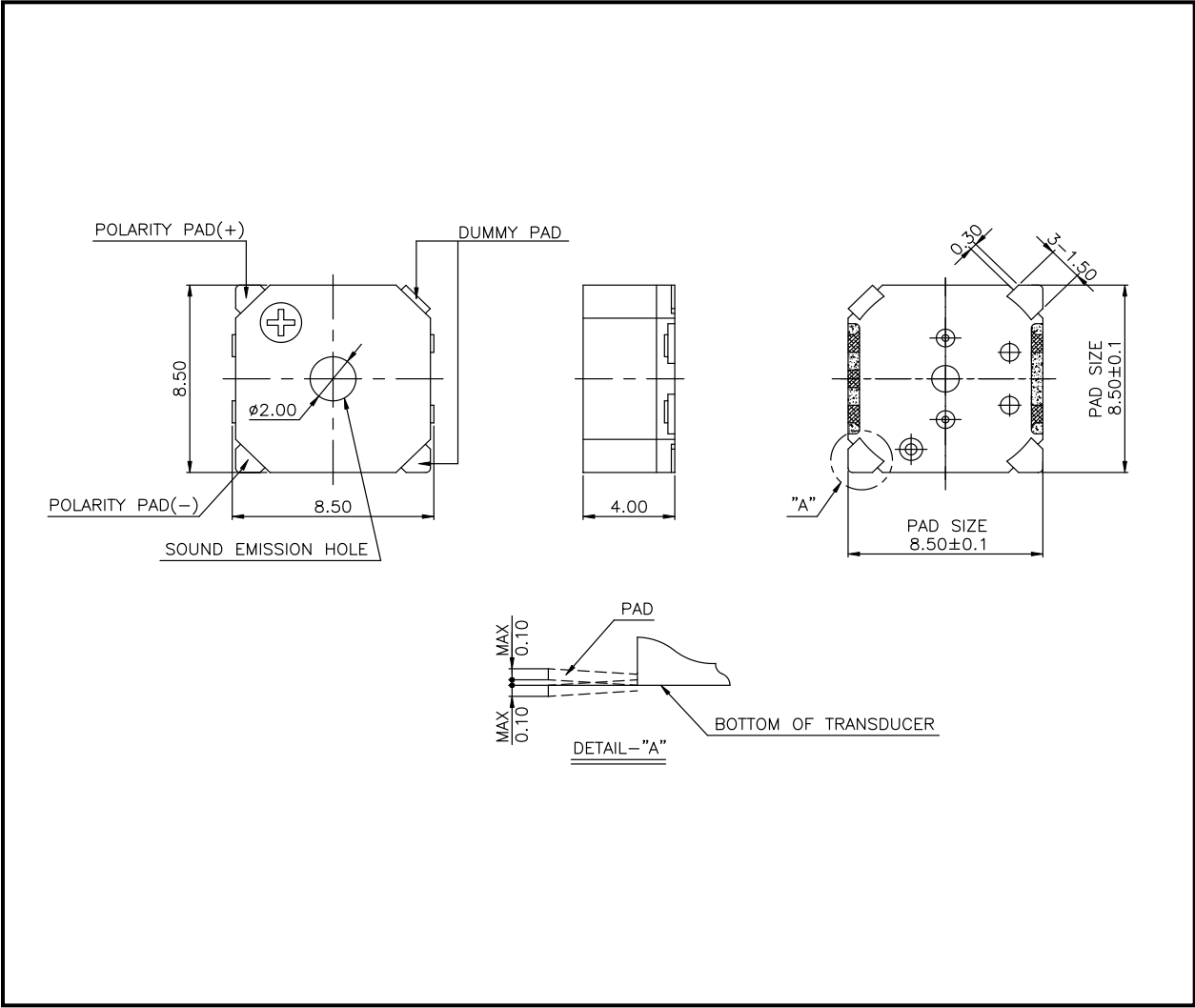
- ① Polarity Mark (+)
  - ② Maker Name
  - ③ Model No.
  - ④ Data code
- The first digit is the number of the year the last digit is the month  
A- January through L – December

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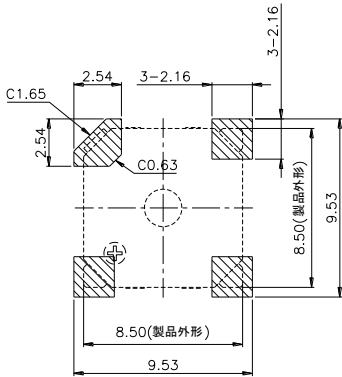
9. Dimensions & Mechanical Layout

9-1. Dimension

|                  |           |
|------------------|-----------|
| Tolerance ; ±0.2 | Unit ; mm |
|------------------|-----------|



. 9-2. Recommend Pattern Land

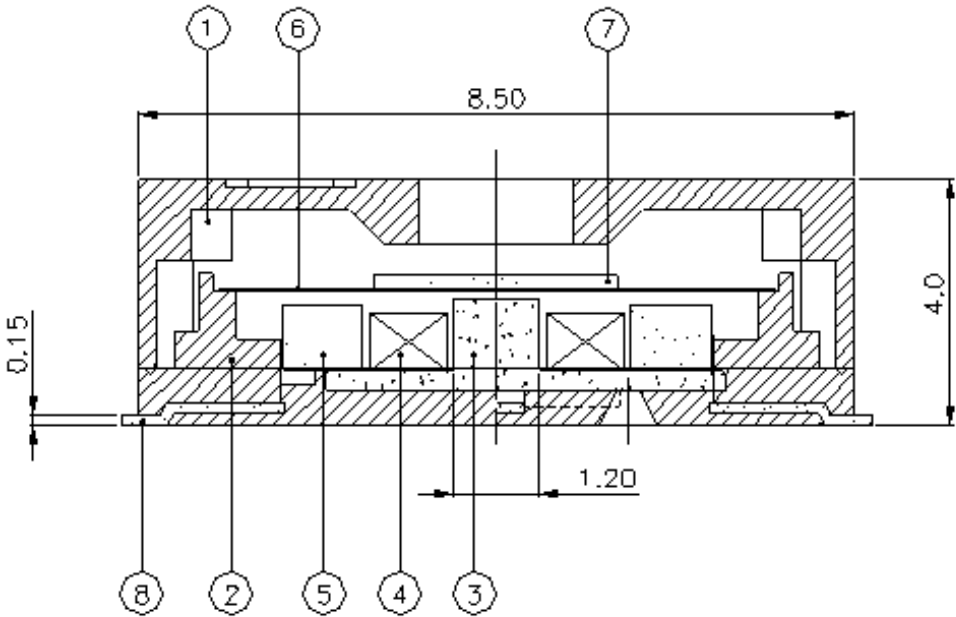


|           |
|-----------|
| Unit ; mm |
|-----------|

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9-3. Structure of TE-8540BYT3H2

| Tolerance |      |
|-----------|------|
| ~1        | 0.03 |
| 1~5       | 0.05 |
| 5~10      | 0.05 |
| 10~20     | 0.1  |
| 20~30     | 0.1  |
| 30~       | 0.15 |

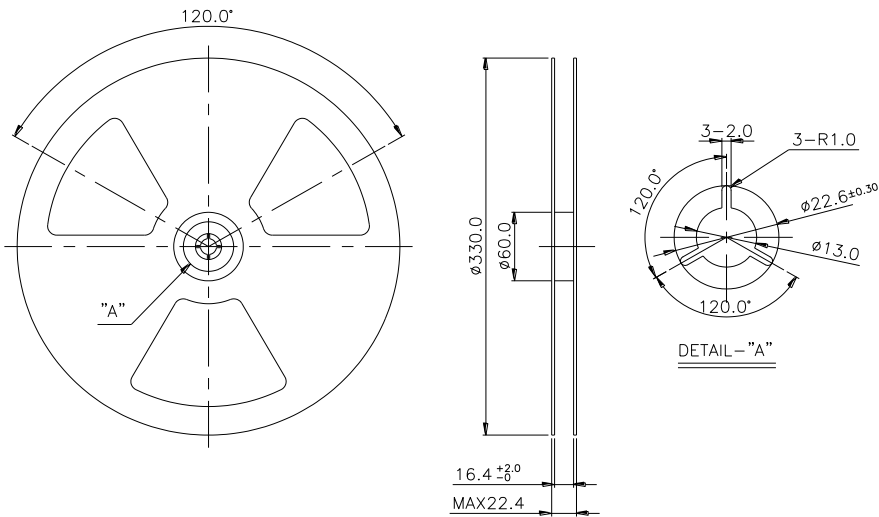


| No | Part Name    | Code No.  | Material  | Remarks     |
|----|--------------|-----------|-----------|-------------|
| 1  | Upper case   | 11990206  | LCP       |             |
| 2  | Lower case   | 110C301   | LCP       |             |
| 3  | Yoke         | 11YK121C  | Pure-Fe   |             |
| 4  | Coil         | 11LW09011 | 3DS-FC/FY |             |
| 5  | Magnet       | 11MG10A9  | Sm-Co     | Mass 0.085g |
| 6  | Resonator    | 1943310A  | PERMALLOY |             |
| 7  | Magnet plate | 11MP1324  | PERMALLOY |             |
| 8  | Terminal     | 110C6100  | PBS       |             |

|                 |                       |          |                  |
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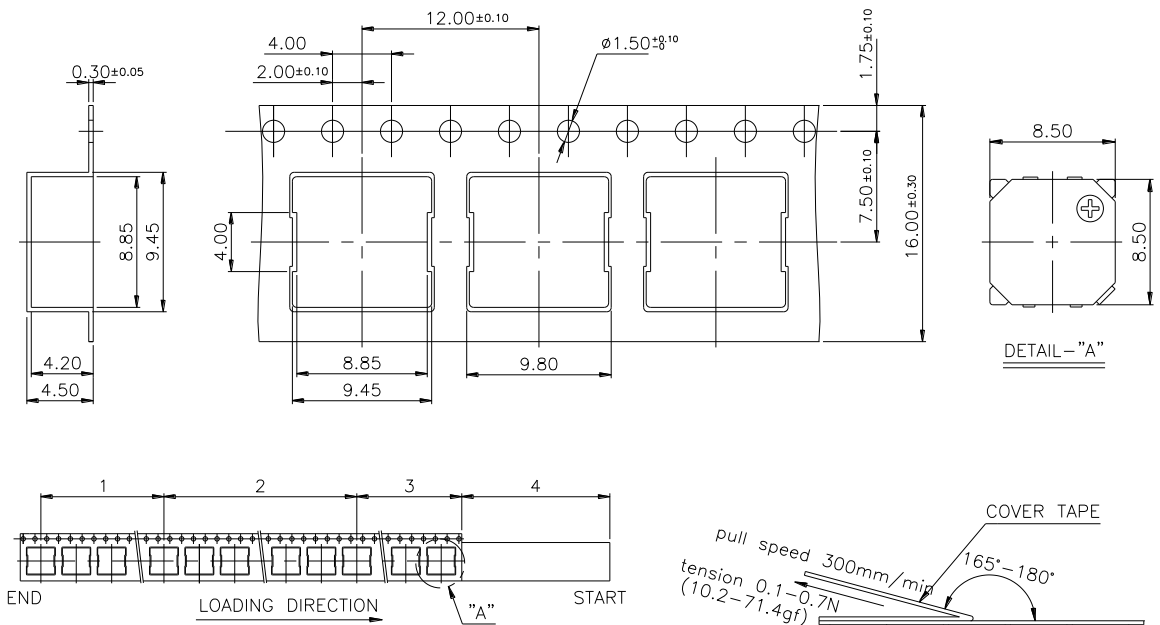
10. Taping Specifications

10-1. Reel Tape

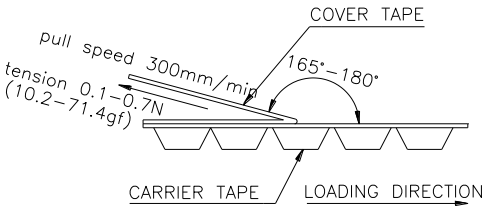


|                       |           |
|-----------------------|-----------|
| Tolerance ; $\pm 0.2$ | Unit ; mm |
|-----------------------|-----------|

10-2. Carrier Tape



1. BARE EMBOSS AREA : 400mm OVER(35PCS OVER)
2. CHIP MOUNT AREA : 1,000PCS
3. BARE EMBOSS AREA : 400mmOVER(35PCS OVER)
4. LEADER AREA : 100mm OVER

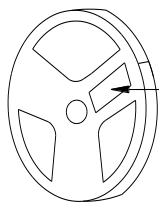


|                       |           |
|-----------------------|-----------|
| Tolerance ; $\pm 0.2$ | Unit ; mm |
|-----------------------|-----------|

|                 |                       |          |                  |
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11. Packing Specifications

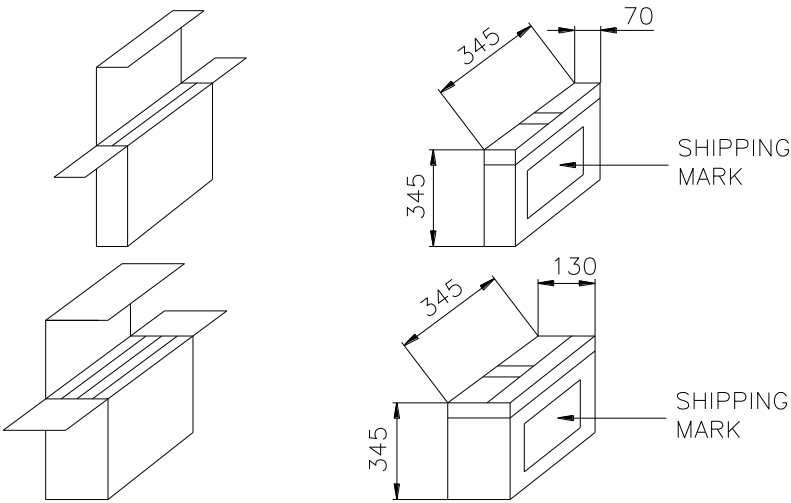
11-1. 1,000 PCS per 1roll by using the tapping M/C.



LABEL →

|            |  |
|------------|--|
| P / O NO.  |  |
| MODEL NO.  |  |
| DATE CODE. |  |
| LOT NO.    |  |
| PART NO.   |  |
| QUANTITY.  |  |
| DATE.      |  |
| VENDOR     |  |

11-2. 2rolls or 4rolls per 1 double(thick) carton box packed by polypropylene tape.



11-3. Content of the label

- Model No.
- Lot No.
- Buyer Parts No.
- Quantity
- Data
- Vendor

11-4. The label to be stick

|           |               |
|-----------|---------------|
| P/O No.   |               |
| Model No. |               |
| Lot No.   |               |
| Part No.  |               |
| Quantity  | pcs           |
| Date      | 201 . . . . . |
| Vendor    | .             |

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12. FIT of TE-8540BYT3H2

1. Reliability =  $R(t) = e^{-\text{Failure rate} \times \text{Operating time}}$   
                   =  $e^{-1/\text{MTBF} \times \text{Operating time}}$
2. Defect of 1pcs is occurred among 100pcs if TE-8540BYT3H2 is operated for 1,000 hours continuously.  
 => **Reliability = R(t) =0.99**  
 therefore,  $0.99 = e^{-\text{Failure rate} \times \text{Operating time}} = e^{-1/\text{MTBF} \times \text{Operating time}}$   

$$\text{MTBF} = \frac{-\text{Operating time}}{\ln 0.99} = \frac{-1000}{\ln 0.99} = \underline{99,499.2 \text{ MTBF}}$$
3. FIT  
       =  $1000000000 / ( \text{MTBF} \times 365 \times 24 )$   
       =  $1000000000 / ( 99,499.2 \times 8760 )$   
       = 1.147298181 FIT

13. Environmental Requirement : This product shall comply with RoHS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |          |                  |
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| TECHTUIT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SPECIFICATIONS        | Page     | 16 / 16          |
| Magnetic Buzzer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MODEL : TE-8540BYT3H2 | Spec. No | TPI-110310-13B15 |
| <p><b>14. Handle with care ( Magnetic buzzer is electronic parts, please follow the following matters.)</b></p> <ol style="list-style-type: none"><li>1. Please do not shock physically, give pressure to the product itself, and also not give stress to the bottom of terminal.</li><li>2. Product characteristic can be effected by surrounding environment such as baffle, cavity and sound insulation etc.</li><li>3. When designing, please consider dust, drop of water, flux smoke and any foreign material.<br/>If the above things enter sound opening parts, it can change product characteristic.</li><li>4. Please make sure product to be used within the parameters of voltage range of operation at stabilized voltage<br/>Rated voltage and standard frequency are desirable.<br/>Sound pressure in specification is value of rated voltage and standard frequency.</li><li>5. Operation of decreasing voltage at series resistance would be insecurity. Please avoid the situation.</li><li>6. Since electric current flow can be 2 to 3 times than average consumption current on peak.<br/>Please prepare your electronic circuit well for supply current.</li><li>7. Regarding transition time from off to on, it should be less than 10 msec for mechanical operate contact ( relay, switch and etc) and less than 1 μsec for contactless (transistor, FET and etc) to set up.</li><li>8. Storage/ operating environment should be normal temperature, 5-25℃ and humidity, 40-60% which does not cause any condensation with sudden change in temperature.</li><li>9. As product includes magnet inside, please do not get closed anything made of metal or has magnetic nature to it. Characteristic value can be changed.</li><li>10. Voltage range of operation is only for that product can be used without breakdown and not for guarantee of voltage, consumption current and characteristic.</li><li>11. The warranty period is for 6 months after produced.<br/>( It does not mean that it will be NG after 6 months, please apply products within 1 year as standard.)</li></ol> <p>Please make sure to store them with said the above normal temperature and humidity.</p> <p>Notices : The change about the material and structure applies in advance, and a change about a process may be made without an application within limits which satisfy the standard of specifications</p> |                       |          |                  |